

Mykhailo Kapustin

Principal Backend Engineer (Python/AI)

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Summary

Backend- and AI-focused engineer with 10+ years of experience, comfortable working across the full stack and with cloud infrastructure (AWS/GCP/Azure) when needed. The last 2–3 years have focused on production LLM/multi-agent systems and graph-based knowledge representation — including architecting and owning a multi-tenant agent platform end to end. EU citizen, Berlin-based, open to full-time, contract, or freelance engagements.

Skills

AI / ML	LangChain, LangGraph (3 yrs, production), knowledge graphs, RAG, multi-agent systems, Google ADK, Vertex AI, Gemini, OpenAI/GPT-4, PyTorch
Languages	Python, TypeScript/JavaScript, SQL
Backend	Python/FastAPI, Node.js, GraphQL, tRPC, WebSockets
Frontend	Vue.js, Nuxt.js, Pinia, React
Cloud / Infra	Docker, Terraform, CI/CD, AWS, GCP, Azure
Databases	PostgreSQL, Redis, MongoDB, Pinecone

Experience

Co-Founder & CTO — ASRP (Advanced Scientific Research Projects, US- and KZ-based holding) Apr 2023 – Present

- Built a transatlantic research holding from scratch: 10+ products across EdTech, AI, media, and blockchain
- Built the first prototype of a LangChain-based forecasting platform (GFS) at an AI hackathon in Berlin (Theta EuroCon)

Selected client engagement (delivered through ASRP)

Multi-tenant AI-organizations platform — a German AI SaaS company

- Designed and built the Python/FastAPI backend for dynamic agents — tool registries, tenant-safe execution, agent scheduling — alongside a NestJS + React + PostgreSQL customer-facing layer (row-level security for tenant isolation)
- Agents read and write into a shared context layer rather than isolated prompts — the same "raw communications → structured relationship data" problem this role works on
- Natural-language agent authoring: describe an agent in plain text, get a working, deployed agent back in ~1 minute
- Top contributor (~45%) across both core repositories; owned the staging → production release flow

Earlier roles (2016–2023): Team Lead / Senior Backend Developer (MOB.325 / LAB325, Ukraine — Node.js, GraphQL, Docker, AWS, ClickHouse), Team Lead (Coelix, Tel Aviv, remote), Senior Backend Developer (Provectus, San Francisco), Full Stack Developer (Ephyros, HYS Enterprise, Top Agent, PHP-academy), Backend Developer (SuperMediaAds) — PHP / Laravel / Vue foundation years.

Publications

11 technical articles on AI agents, infrastructure, and NLP — ~124K combined reads across Habr, Medium, and Hashnode. Selected:

- [Docling in Working with Texts, Languages, and Knowledge](#) — 13.9k views. Documents as graphs of typed, linked nodes; LLM-driven graph querying.
- [Building AI Agent Infrastructure](#) — 10k+ views. Filesystem Registry → Package Manager → AgentOps/MCP&A2A — scaling multi-agent systems.
- [Google ADK and "Startup Technical Guide: AI Agents"](#) — 5.3k views. Grounding: RAG → GraphRAG → Agentic RAG for production-reliable agents.
- [Building a Resume Matcher with tRPC, NLP, and Vertex AI](#) — 24k+ reads. Structured, predictable JSON output from an LLM instead of free text.

Education

Master's, Automation and Computer-Integrated Technologies — State University of Intelligent Technologies and Telecommunications (2017–2022)

Work Authorization

EU citizenship — no visa/work-permit sponsorship required in the EU. Berlin-based. Open to full-time, contract, or freelance engagements — also able to contract directly through US- and Kazakhstan-based entities where useful.